LESSON 5.6

Represent Three-Dimensional Figures in Two Dimensions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of a solid by using its net.

WHY THIS MATTERS Your class is designing custom wrapping paper for time capsule boxes. To figure out how much paper you need, you have to find the total surface area of each box. The trick? Unfold the box into a net so you can see every face laid flat! One time capsule box is 6 inches long, 4 inches wide, and 2 inches tall. How much wrapping paper covers the whole box?

REMEMBER

Represent Three-Dimensional Figures in Two Dimensions

Representing 3D Solids with Nets Surface area is the total area of all the flat faces of a solid. A net unfolds the box flat so you can see and add all 6 faces. The answer uses square units.

1. A net is a 2D flat pattern that folds into a 3D solid without overlapping.
2. Rectangular prism net has 6 rectangular faces in 3 congruent opposite pairs.
3. Surface Area is the total sum of the areas of all flat faces on the net

Words: Surface area *(Área de superficie)* · Net *(Plantilla (desarrollo plano))* · Face *(Cara)* · Two-dimensional *(Bidimensional)* · Edge *(Arista)*

1 **Circle the letter of the best answer. Show how you know.**

What is the volume of a rectangular prism with l = 4 in, w = 3 in, h = 5 in?

(A) 12 in³ (B) 24 in³

(C) 60 in² (D) 60 in³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A box has a base area of 12 square cm and a height of 3 cm. What is its volume?

(A) 4 cm³
(B) 15 cm³
(C) 36 cm³
(D) 36 cm²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A box has a volume of 36 cm³. Its length is 6 cm and width is 3 cm. What is the height?

(A) 2 cm
(B) 3 cm
(C) 6 cm
(D) 12 cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

How is volume different from area?

(A) Volume uses cubic units; area uses square units
(B) Volume uses square units; area uses cubic units
(C) Volume only applies to cubes; area applies to all shapes
(D) There is no difference

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A gift shop wraps boxes with decorative paper. A box is 14 inches long, 10 inches wide, and 6 inches tall. One sheet of wrapping paper covers 400 square inches. What is the surface area of the box?

(A) 280 in²
(B) 284 in²
(C) 568 in²
(D) 840 in²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

A gift shop wraps boxes with decorative paper. A box is 14 inches long, 10 inches wide, and 6 inches tall. One sheet of wrapping paper covers 400 square inches. Why is each face pair multiplied by 2?

(A) A rectangular box has two identical faces of each size
(B) Because you always double in geometry
(C) Because there are 2 sheets of paper
(D) To convert to square inches

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The total surface area is 158 in². Which face pair contributes the most area? Why does that make sense when you look at the dimensions?

SENTENCE STARTER The ___ faces contribute the most area (___ in²) because their dimensions ___ × ___ = ___ are the largest. The total SA is ___ + ___ + ___ = ___ in².

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each solid to the net it folds up from.

_____ Cube	(A) Net of 6 equal squares
_____ Rectangular prism	(B) Net of 6 rectangles (3 matching pairs)
_____ Triangular prism	(C) Net of 2 triangles and 3 rectangles
_____ Square pyramid	(D) Net of 1 square and 4 triangles
_____ Faces on any rectangular prism	(E) 6 faces
_____ Square units (in², cm²)	(F) How surface area is measured

2 **Write the name of the correct group on each line.**

Sort each flat shape: is it a valid net that folds into a closed time capsule box, or not a complete net?

Groups: **Valid net (folds into a closed box)** **Not a complete net**

_____ Six rectangles arranged in a cross shape

_____ Six rectangles in a T shape (3 matching pairs)

_____ Four rectangles in a straight row (top and bottom missing)

_____ A single rectangle by itself

_____ Two squares and four rectangles for a box

_____ Seven rectangles (one extra face)

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A gift shop wraps boxes with decorative paper. A box is 14 inches long, 10 inches wide, and 6 inches tall. One sheet of wrapping paper covers 400 square inches. How many sheets of wrapping paper do you need?

(A) 1
(B) 2
(C) 3
(D) 568

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

You unfold a time capsule that is a cube. Its net shows 6 equal squares, and each square has an area of 4 cm². What is the cube's total surface area?

(A) 10 cm²
(B) 12 cm²
(C) 16 cm²
(D) 24 cm²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 **Circle the letter of the best answer. Show how you know.**

Your team unfolds a rectangular time capsule box into a flat net. How many faces (flat sides) will the net show in all?

(A) 4
(B) 5
(C) 6
(D) 8

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A time capsule box measures 2 in × 3 in × 4 in. Using its net, you add up all six faces. What is the total surface area?

(A) 24 in²
(B) 26 in²
(C) 48 in²
(D) 52 in²

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

The total surface area is 158 in². Which face pair contributes the most area? Why does that make sense when you look at the dimensions?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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